

Seminar Heyting-Valued Types

1 CwFs

2. Dependency in functors

Let $P: \mathcal{P} \rightarrow \mathcal{C}$. Formally a dependent functor over $\mathcal{P}$ is a functor $Q: \mathcal{P} \rightarrow \mathcal{D}$.

Category $\mathcal{P}$:

$ob: (\mathcal{C}, \mathcal{P}); C: \mathcal{C} \rightarrow \mathcal{P}; PC$

$mor: (H, P): (C', P.f) \rightarrow (C, P); f: \mathcal{C}(C', C); P: PC$

$Q: (\mathcal{C}: \mathcal{C}) \rightarrow (\mathcal{P} \rightarrow \mathcal{D})$

$Q: \{C: \mathcal{C}\} \rightarrow \{C: \mathcal{C}\} \rightarrow \{P: PC\} \rightarrow \mathcal{D}(QCP, QCP.f)$

$QC: PC \rightarrow \mathcal{D} = Fam_{\mathcal{D}}(PC)$

$Qf: (P: PC) \rightarrow \mathcal{D}(QCP, QCP.f)$

$(\Phi_f, Qf): QC \rightarrow QC'$

write $Psh \mathcal{P}$ for $(\mathcal{P}^{op} \rightarrow Set) \rightarrow Psh(\mathcal{P})$

1. Fix a Cat. of contexts Ctx with ε emp. ctx.

Idea:

→ Have Types over $Ctxs$

→ Have Terms over Types

→ subst, coh, and strictness.

Types and terms as presheaves give subs.

Problem: "What is a presheaf over Types?"

3. CwF

$-Ctx: Cat. \text{ of term. } \Gamma, \Delta, \dots$

$-Ty: Psh \ Ctx, \quad A, B, \dots \quad Ty \ x := -[x]$

$-Tm: Psh \ Ty, \quad a, b, \dots \quad Tm \ x := -[x]$

$- \vdash := \vdash_{Ty} \rightarrow Ctx. \quad \Gamma, A$

$- \vdash := \vdash_{\Gamma} \ x: \Delta \rightarrow \Gamma. \quad Tm \ \Delta \ A[x] \cong Ctx(\Delta, P.A) \vdash \varphi$

$\hookrightarrow (\pi_P^A, var_P^A) = \varphi(id_{P.A}); \pi_P^A: P.A \rightarrow P, var_P^A: Tm \ P.A \ A[\pi_P^A]$

I. $\pi, var = id$

II. $(x.a) \circ j = (x \circ j).a[j]$

III. $\pi \circ (x.a) = x$

$var_P[x.a] = a; Tm \ \Delta \ A[x]$

$var_P[id.a] = a; Tm \ P.A; -[id.a] = -(a)$

$\Delta.A[x] \xrightarrow{\gamma_A} P.A \quad \gamma.A = (\gamma \circ \pi).var$

$\pi \downarrow \quad \hookrightarrow \quad \downarrow \pi \quad \gamma^A$

$\Delta \xrightarrow{\gamma} P$

4. Ex. Set theoretic model

Let F small cat w/ terminal obj.

$-Ctx := Set$

$-Ty \ P := Fam_F \ P = P \rightarrow Ob \ F$

$-Tm \ P.A := \prod_{g: \Gamma. F(1, A(g))} = \prod_{g: \Gamma. A(g)}$ subby precdp.

$- \Gamma.A := \sum_{g: \Gamma. F(1, A(g))} = \sum_{g: \Gamma. A(g)}$

$- \gamma.a := \prod_{d: \Delta. (x(d), a(d))}$

$\gamma: \Delta \rightarrow \Gamma$

$a: Tm \ \Delta \ A[\gamma] = \prod_{g: \Delta. F(1, A(x(g)))}$

Set theoretic model is for $F = U$; U univ. ob. sets.
i.e. k -small for some inaccessible κ .

5. Constructions

We wish to calculate things so extensional.

$- \Pi type: \Gamma \vdash A, \Gamma \vdash B$

$- \Pi A B: Ty \ P; (\Pi A B)[x] = \prod_{A[x]} B[x.A]$

$- \vdash: Tm \ \Gamma \ (\Pi A B) \cong Tm \ \Gamma.A \ B: \lambda_{\Gamma}^{A \vdash B}$

$- Tm \ x \ \Pi A B = Tm \ x.A \ B$

$\hookrightarrow \alpha_P \beta a = \hat{f}(a): Tm \ \Gamma \ B(a)$

$\alpha_P: Tm \ \Gamma \ \Pi A B \xrightarrow{\sim} Tm \ \Gamma \ A \rightarrow Tm \ \Gamma \ B(a)$

$- \Sigma type:$

$- \Sigma A B: Ty \ P \ (\Sigma A B)[x] = \sum_{A[x]} B[x.A]$

$- pair: \Gamma.A.B \xrightarrow{\sim} \Gamma.\Sigma A B; \pi^{\Sigma A B}_{pair} = \pi^A \circ \pi^B$

$- \gamma \Sigma A B \circ pair = pair \circ \gamma.A.B$

Note: This is Extensional. In general $pair$ is not iso and we define an elim.

$- Unit:$

$- \mathbb{1}: Ty \ \varepsilon; \pi_{\varepsilon}^{\mathbb{1}}: iso \ \varepsilon \rightarrow \mathbb{1} \in \varepsilon$

$- NNO$

$- N: Ty \ \varepsilon; \varepsilon.N$ is NNO (parametrised)

$- Id_{ent}:$

$- Id_A: Ty \ P.A.A^0, Id_A[x] = Id_A[x]$

$- refl: \Gamma.A \xrightarrow{\sim} \Gamma.A.A.Id_A$

6. Ex. Set theoretic model

$- \Pi type: \gamma \mapsto \prod_{a: A(\gamma)} B(\gamma, a)$

$- Tm \ \Gamma \ \Pi A B \cong \prod_{g: \Gamma. A(g)} B(g, g)$

$\cong \prod_{g: \Gamma. A(g)} B(g, g)$

$-(\Pi A B)[x] = \prod_{a: A(x)} B(x, a)$

$= \prod_{A[x]} B[x.A]$

$- \Sigma type:$

$- \Sigma A B: \gamma \mapsto \sum_{a: A(\gamma)} B(\gamma, a)$

$- pair(\gamma, a, b) = (\gamma, a, b)$ is iso

$- Unit: \varepsilon \mapsto \{*\}. \varepsilon.\mathbb{1} = \sum_{\varepsilon} \{*\} = \{*\}$

$- NNO: \varepsilon \mapsto \mathbb{N}$

$- Id: Id_A(\gamma, a, a') = \{*\} \mid a = a'$

$- refl(\gamma, a) = (\gamma, a, a)$

2 Heyting-valued Types

$\mathcal{C}: ob \ A(A, R); A \in Set, R: A \times A \rightarrow \Omega$

$\Omega_1 - R(a, a') \leq R(a', a)$

$\Omega_2 - R(a, a') \wedge R(a', a'') \leq R(a, a'')$

$mor \ f(A, R) \rightarrow (B, Q): f: A \times B \rightarrow \Omega$

$\Omega_1 - f(a, b) \leq f(a', b')$

$\Omega_2 - R(a, a') \wedge f(a, b) \wedge R(b, b') \leq f(a, b')$

$\Omega_3 - f(a, b) \wedge f(a, b') \leq b = b'$

$\Omega_4 - \exists a \leq \bigvee_{b \in B} f(a, b)$

$R := A$

$R(a, a') = \exists a$

$A = (A, R)$

$f: A \rightarrow B$

CwF:

$-Ctx := \mathcal{C}$

$-Ty \ P := \{A: \mathcal{P} \mid \exists ob \ \mathcal{C} \mid \exists g \neq g' \leq \bigvee \{T \mid B(g) = B(g')\}\}$

$Ty \ x: \Delta \rightarrow P \mapsto (d \mapsto A(x(d)))$

$-Tm \ P.A := \{g: \Gamma. A(g)\} \xrightarrow{\sim} \Omega \mid M1-M4$

$Tm \ x.A := f \mapsto (d, a) \mapsto f(x(d), a)$

$- \Gamma.A := \{g: \Gamma. A(g), g \neq g' \wedge a \neq a'\}$

$- \gamma.a := (d, g, a) \mapsto x(d, g) \wedge a(d, a')$

$\gamma: \Delta \rightarrow \Gamma$

$a: Tm \ \Delta \ A[\gamma] = \{ \sum d: \Delta. A[x](d) \rightarrow \Omega \mid M1-M4 \}$

$- \Pi type:$

$- \Pi A B(g) = (\{ \sum a: A(g). B(g, a) \rightarrow \Omega \mid M1-M4 \}, \wedge_{a \in A} f(a) = g(a))$

$- Tm \ \Gamma \ \Pi A B = Tm \ \Gamma.A \ B$ (check cond)

$- \Sigma type:$

$- \Sigma A B(g) = (\{ \sum a: A(g). B(g, a), a = a' \wedge b = b' \})$

$- pair(g, a, b) = (g, a, b)$ is iso

$- Unit: \varepsilon \mapsto \{*\}, T$

$- NNO: \varepsilon \mapsto (\mathbb{N}, \bigvee \{ \vdash \mid n = m \})$

$- Id_A: (g, a, a') \mapsto \{*\} \mid a = a'$

$refl(g, a) = (g, a, a)$ is iso